



TECHNOLOGY INFORMATION IN ADMINISTRATION OFFICE

Group 2

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Gratitude is extended to:

The lecturers/teachers who have provided direction and motivation in the preparation of this paper. This paper is realized to be far from perfect. Therefore, constructive criticism and suggestions are highly appreciated for the improvement of future papers. It is hoped that this paper can provide benefits and add insight to readers, especially regarding the use of information technology and computers in administrative activities.

BACKGROUND

The development of Information and Communication Technology (ICT) in today's digital era has had a significant impact across various fields, including education. One of the most affected aspects is school administration. The use of computer technology in school administration is no longer considered optional but has become a necessity to support the smooth management of data and educational services.

School administration involves various activities such as student data recording, grade management, document archiving, report preparation, communication with parents, and school financial management. In the past, most of these tasks were carried out manually using paper and physical files, whereas computers now provide a system that is faster, more accurate, and more integrated. For instance, the use of school information systems (SIS) or school management systems enables teachers, administrative staff, and principals to manage data more effectively.

In addition to offering many benefits, the use of computers in school administration also requires digital literacy skills among education personnel. Other challenges include the need to ensure the security of student and teacher data, the availability of adequate devices, and the regular updating of software. With the integration of computer technology in school administration, the educational process is expected to become more organized, transparent, and efficient.

SYSTEM ANALYSIS

System Summary:

School administration is an essential part that functions to support the smooth implementation of teaching and learning activities. The management of student data, teacher data, grades, finances, and correspondence requires a system that is fast, accurate, and well-integrated. Therefore, the implementation of Information and Computer Technology in school administration is highly necessary. This system is designed to simplify the work of administrative staff, teachers, and school principals in accessing and managing information efficiently.

To support the school's vision and mission in providing better educational services, a reliable computer network is needed. Networking hardware such as routers and switches are the main components. Routers are used to connect the school's local network (LAN) with wider networks such as the internet, and to manage data traffic so that information can reach its destination through the best possible route. Meanwhile, switches play a role in managing data traffic between devices within the school's network, ensuring that each computer used by staff, teachers, and administrative units can be interconnected without data collisions.

From the software side, school administration computers use modern operating systems such as Windows 11, which support stability and security. For work productivity, Microsoft 365 applications are also used, providing cloud-based services such as Word, Excel, and PowerPoint. In addition to Microsoft applications as the primary software, there are also supplementary programs to enhance stability, including CCTV applications, scanner applications, CorelDRAW, E-Learning platforms, and Zoom. These applications help administrative staff in creating reports, storing data, and collaborating in real-time. The system also enables cloud-based data storage, which reduces the risk of data loss due to device failure.

In addition, data security is also a top priority. Student and teacher data are protected through the use of authentication systems on computers, regularly updated antivirus software, and firewalls that help prevent external threats. For physical security, schools may also install CCTV in administrative offices and other important areas to ensure a safe and comfortable environment.

HARDWARE SPECIFICATION



Figure I Monitor Samsung Odyssey LS24DG302EEXXS (REF:
<https://www.samsung.com/my/monitors/gaming/odyssey-g3-g30d-24-inch-180hz-freesync-ls24dg302eexxs/?>)

Specification of Hardware

Resolution	FHD (1,920 x 1080)
Aspect Ratio	16:9
Brightness	(Typical) 250cd/m ²
Contrast Ratio Static	3,000:1(Typ.)
Response Time	1ms(MPRT)
Refresh Rate	Max 180Hz
Viewing Angle (H/V)	178°(H)/178°(V)



Figure II Smart Wireless Keyboard VG-KBD1000 (REF:
<https://www.tokopedia.com/rekomendasi/10213050378>)

Specification of Hardware

Connection type:	Bluetooth 2.1
Power	2 x AA Batteries
Dimensions (unit)	317 x 123.8 x 9.6 mm
Weight	445 g
Key type	Pantograph



Figure III Gaming Mouse PC RGB PREDATOR Redragon M612 RGB (REF: <https://shopee.co.id/product/250151994/10508263494?>

Specification of Hardware

DPI range	500-8,000
Body Material	Frosted Coating with textured rubber grips on the left side
Acceleration	15G
Weight	135 g
RGB lighting	5 RGB backlight



Figure IV 4. Epson L5290 Printer (REF: <https://www.tokopedia.com/rekomendasi/6099695375>)

Specification of Hardware

Max print resolution	5760 x 1440 dpi
Print speed	Up to 33 paper/minute
Connectivity	Wi-Fi, Ethernet, USB
Ink System	Refillable EcoTank
Dimensions	375 x 347 x 237 mm
Weight	5 kg



Figure V Hikvision Turbo HD Turret Camera DS-2CE76D0T-EXLPF (REF: <https://www.monotaro.id/items/s037427450.html>)

Specification of Hardware

Resolution	2 MP (1080p)
Lens	2.8 mm or 3.6 mm
Night vision	IR up to 20 m, White light up to 20 m
Power	12 V DC, ~3.5 W
Video output	TVI, AHD, CVI, CVBS (switchable)



Figure VI PROLiNK Wireless Dual and Gigabit Router PRC3801 AC 1200 MU-MIMO (REF: <https://www.monotaro.id/items/s010002681.html>)

Specification of Hardware

Power	12 V DC, ~ 10 W max
WiFi standard	802.11ac (WiFi 5), dual-band (2.4 & GHz)
Max speed	300 Mbps (2.4 GHz)
Ports	1 x Gigabit WAN, 4 x Gigabit LAN



Figure VII TP-Link 24-Port Gigabit Desktop Rackmount Switch TL-SG1024D (REF: <https://www.tokopedia.com/ganeshazone/tp-link-24-port-gigabit-desktop-rackmount-switch-tl-sg1024d-switch-tp-link-24-port-gigabit-1730831885645481207?>)

Specification of Hardware

Ports	24 x RJ-45 (10/100/1000 Mbps)
Power (max)	~13,6 W
Forwarding Rate	35.6 Mbps
Dimensions	294 x 180 x 44 mm



Figure IX CPU HP PRODESK 600 G1 CORE I3 4150 RAM 4GB SSD 128GB (REF: <https://shopee.co.id/product/15139273/20232359209?>)

Specification of Hardware

Ports	24 x RJ-45 (10/100/1000 Mbps)
Power (max)	~13,6 W
Forwarding Rate	35.6 Mbps
Dimensions	294 x 180 x 44 mm



Figure XI WiFi IP Phone Flying Voice FIP11CP - Color LCD & Support PoE (REF: shopee.co.id/product/301674747/28379413507)

Specification of Hardware

Display	2.4" color LCD (320×240) with backlight
Connectivity	Built-in 2.4 GHz WiFi
Network	Dual Ethernet ports (10/100 Mbps)
Audio	HD voice support via G.722 and Opus codecs



Figure XII HiLook DVR-204G-M1 (REF: <https://tk.tokopedia.com/ZSDqcmXvx/>)

Specification of Hardware

Channels	4CH (supports 2MP cameras)
Video Input	HDTVI / AHD / HDCVI
Video Output	HDMI, VGA
Storage	1 SATA (up to 4TB HDD)
Network	1 RJ45 (10/100 Mbps)
Power	12V DC, ~10W



Figure XIII KOLMI Dual Way Window Counter Intercom (REF: <https://tk.tokopedia.com/ZSDq3evMf/>)

Specification of Hardware

Material	Aluminum alloy panel
Working voltage	12 V DC / 500 mA
Microphone sensitivity	-48 dB (20 Hz – 16 kHz)
Audio	2 W + 2 W (Outside + Inside)
Weight (main unit)	350 g



Figure XIV Samsung F27T450FQN – 27" IPS Monitor (REF: <https://id.shp.ee/vxCSFYU>)

Specification of Hardware

Screen / Panel	27" IPS, Flat
Resolution	1920 × 1080 (Full HD)
Inputs / Connectivity	2 × HDMI 1.4, 1 × DisplayPort 1.2
Brightness (typical)	250 cd/m ²
Color Support	16.7M colors
Refresh Rate	75 Hz



Figure XV DP to HDMI Converter Adol GD-ED058 ASUS (REF: <https://www.blibli.com/p/dp-to-hdmi-converter-adol-gd-ed058-asus-license-garansi-24-bulan-support-4k-uhd-30hz/ps--ADI-70065-00009>)

Specification of Hardware

Type	DisplayPort (male) to HDMI (female) converter (one-way)
Length	0.15 m (15 cm)
Max Resolution	4K @ 30 Hz



Figure XVI Wilson Electronics RG-6 Low-Loss Coaxial Cable (REF: <https://id.shp.ee/dqv8HPu>)

Specification of Hardware

Type	75 Ω coaxial cable (RG-6 standard)
Center Conductor	18 AWG solid copper-clad steel
Attenuation (Loss)	~0.83 dB per 10 ft @ 800 MHz; ~1.35 dB per 10 ft @ 1900 MHz



Figure XVII UGREEN Kabel Power Printer AC 250V 50607 (REF<https://id.shp.ee/ptJNFGf>)

Specification of Hardware

Rated Voltage	250 V
Rated Current	2.5 A
Conductor / AWG	2 × 0.75 mm ² copper



Figure XVIII Vention Cat6 UTP Ethernet Cable
(REF: <https://id.shp.ee/t4b9Qp3>)

Specification of Hardware

Category	Cat6 UTP (Unshielded Twisted Pair)
Max Data Rate	1,000 Mbps (Gigabit Ethernet)
Bandwidth / Frequency	250 MHz

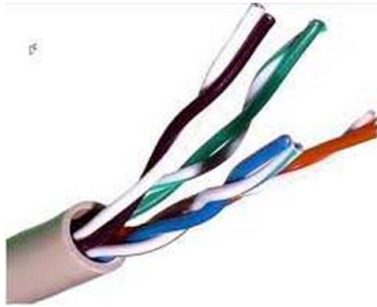


Figure XVIII Kabel TLP ITC 4-core Supreme (<https://www.kmpkabel.com/product/itc-kabel-dan-jelly-armour-5059242>)

Specification of Hardware

Voltage Rating	300V
Conductor Diameter	0.5 mm
Insulation	PVC (Polyvinyl Chloride)



Figure XIX WEBSONG Kabel Fiber Optic SC-SC (REF:<https://www.blibli.com/p/websong-kabel-fiber-optic-sc-sc-50-meter/ps--TOM-25981-00796>)

Specification of Hardware

Length	100 meters
Type	Duplex (2 fibers)
Operating temperature	-20 °C to +70 °C

SOFTWARE SPECIFICATION

1. Operating System

An operating system is the basic software that regulates the work computers in school administration offices. With modern operating system such as Windows 11, computers can run stably, securely, and support various administrative applications such as Microsoft 365, scanners, and E-Learning. In addition to making it easier for staff to manage files, create reports, and access data, the operating system also comes with security features such as user authentication and regular updates to protect school data. Thus, the operating system is the main foundation for all administrative activities to run more efficiently, orderly, and professionally.

2. Microsoft 365

Microsoft 365 is an advanced cloud-based productivity platform that includes applications such as Microsoft Teams, Word, Excel, PowerPoint, Outlook, and OneDrive, as well as intelligent cloud services and advanced security. Start using free web and mobile apps or upgrade to a premium plan to get access to apps, storage, and other features. Microsoft 365 is very helpful for school administration work. Microsoft Word is used to create letters, reports, and official documents quickly and neatly. Microsoft Excel is useful for managing student data, attendance, grades, and financial reports accurately and efficiently. Meanwhile, Microsoft PowerPoint supports presentations in school meetings or social gatherings with attractive and easy-to-understand displays.

3. CCTV Applications

CCTV applications play an important role in maintaining security and order in the school environment, especially in administrative areas. These applications make it easy for staff to monitor activities in real time via computers or other devices, so that potential disturbances or suspicious activities can be detected immediately. With a CCTV system in place, data security, school assets, and employee comfort are better guaranteed, while also creating a safe and orderly administrative environment.

4. Scanner

The scanner application plays an important role in supporting the work efficiency of school administrative staff. With this application, important documents such as decision letters, financial reports, student data, and correspondence archives can be easily converted from physical to digital form. This digitization process makes documents more secure, as it can be stored on computers or in the cloud so that its not easily lost or damaged like paper archives. In addition, scanner applications also facilitate the distribution of information. Scanned documents can be directly shared via email or cloud-based applications with teachers, principals, or external parties such as the education office, without having to duplicate physical documents. This certainly speeds up communication and reduces the cost of using paper. The use of scanner applications in school administration offices can increase staff productivity and create a modern, paperless, and environmentally friendly administration system.

5. Coreldraw

CorelDRAW plays an important role in supporting graphic design needs in school administration offices. This application is used to create various media, such as student and teacher ID cards, school brochures, activity banners, and letters or invitations with a more attractive and professional appearance. With good visual design, information delivery becomes more effective and easier to understand by students, teachers, and parents.

In addition, CorelDRAW also assists in the creation of publication media such as posters of rules and regulations, activity schedules, and socialization materials that require clear visual displays. Not only does it improve the neatness of administration, this application also strengthens the image and identity of the school through the creation of high-quality logos or educational promotional materials. Thus, CorelDRAW contributes to a more modern, creative, and communicative school administration system.

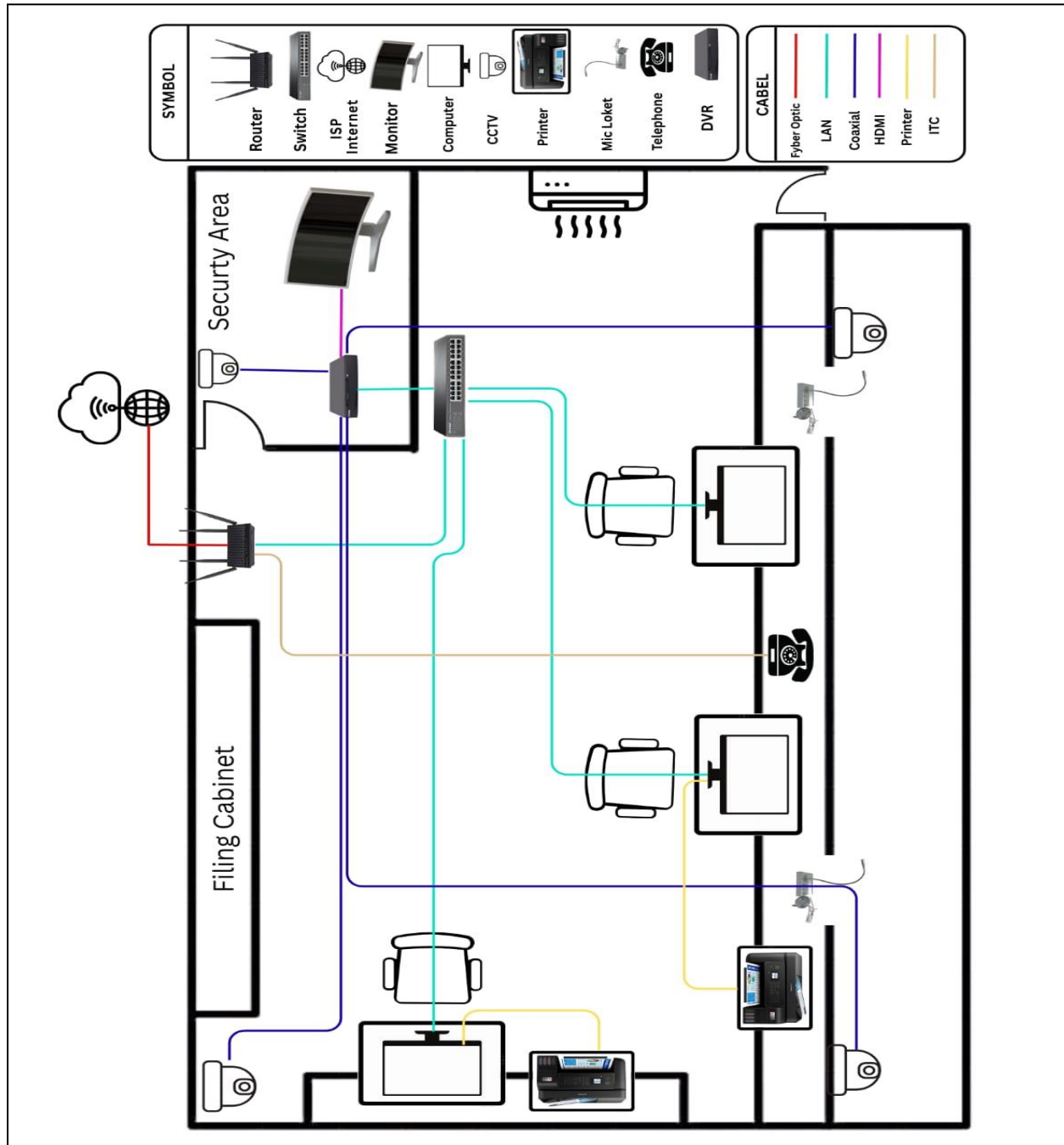
6. E-learning

E-Learning serves as a digital tool to support the learning process and school administration. Through this application, teachers and administrative staff can upload materials, set schedules, store grade data, and interact with students online. This system also facilitates the management of student learning data, which can be accessed at any time without the necessity of physical documents. In addition, E-Learning assists schools in documenting academic activities in an organized and transparent manner, making it easier to compile reports and conduct evaluations. By utilizing E-Learning, schools can provide educational services that are more modern, effective, and in line with technological developments.

7. Zoom

Zoom has become an essential application in supporting long-distance communication in schools. This application is used for internal staff meetings, coordination with teachers, socialization with parents, and online training for educators. Through its video conference feature, communication can be carried out directly and interactively without having to meet physically, thus saving time and costs. Additionally, Zoom makes it easy to document meeting outcomes through its recording feature, which can be used as an archive for administrative purposes. By utilizing Zoom, schools have access to more flexible, efficient, and organized communication.

NETWORK TOPOLOGY DESIGN



The design used in this room is the Star Topology

Reason: All devices (PC, Printer, CCTV, Access Point, etc) are connected to one central device, the switch

- The switch acts as the main connector and controls the data traffic between devices
- And if one cable is cut, only that device is disconnected while the rest of network keeps working

BUDGET

DEVICES USED IN ADMINISTRATION OFFICE

No.	Hardware	Product	Quantity	Price	Cost
1.	Pc	Monitor Samsung Odyssey LS24DG302EEXXS	4	Rp. 2.400.000	Rp. 9.600.000
2.	Keyboard	Smart Wireless Keyboard VG-KBD1000	4	Rp. 699.000	Rp. 2.796.000
3.	Mouse	Gaming Mouse PC RGB PREDATOR Redragon M612 RGB	4	Rp. 260.000	Rp. 1.040.000
4.	Printer	Epson L5290 Printer	2	Rp. 894.000	Rp. 7.788.100
5.	CCTV	Hikvision Turbo HD Turret Camera DS-2CE76D0T-EXLPF	4	Rp. 479.900	Rp.1.919.600
6.	Router	PROLINK Wireless Dual and Gigabit Router PRC3801 AC 1200 MU-MIMO	1	Rp. 689.900	Rp. 689.900
7.	Switch	TP-Link 24-Port Gigabit Desktop Rackmount Switch TL-SG1024D	1	Rp. 1.065.000	Rp. 1.065.000
8.	Cpu	CPU HP PRODESK 600 G1 CORE I3 4150 RAM 4GB SSD 128GB	4	Rp. 1.250.000	Rp. 5.000.000
9.	Telepon	WiFi IP Phone Flying Voice FIP11CP - Color LCD & Support PoE	1	Rp. 898.000	Rp. 898.000
10.	DVR	HiLook DVR-204G-M1	1	Rp. 554.800	Rp. 554.800
11.	Mic & speaker	KOLMI Dual Way Window Counter Intercom	1	Rp. 499.000	Rp. 499.000
12.	Monitor	Samsung F27T450FQN – 27" IPS Monitor	1	Rp. 1.839.080	Rp. 1.839.080

DEVICES USED IN ADMINISTRATION OFFICE

No.	Hardware	Product	Quantity	Price	Cost
13.	CO axial cabel	Wilson Electronics RG-6 Low-Loss Coaxial Cable	1	Rp. 676.200	Rp. 676.200
14.	Power printer cabel	UGREEN Kabel Power Printer AC 250V 50607	1	Rp. 50.000	Rp. 50.000
15.	HDMI cabel	DP to HDMI Converter Adol GD-ED058 ASUS	1	Rp. 64.900	Rp. 64.900
16.	LAN cabel	Vention Cat6 UTP Ethernet Cable	1	Rp. 25.000	Rp. 25.000
TOTAL				Rp.34.505.580	